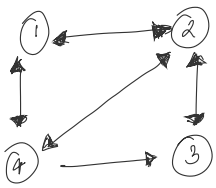


Traveling Salesman Problem

Monday, 9 February 2026 9:58 PM

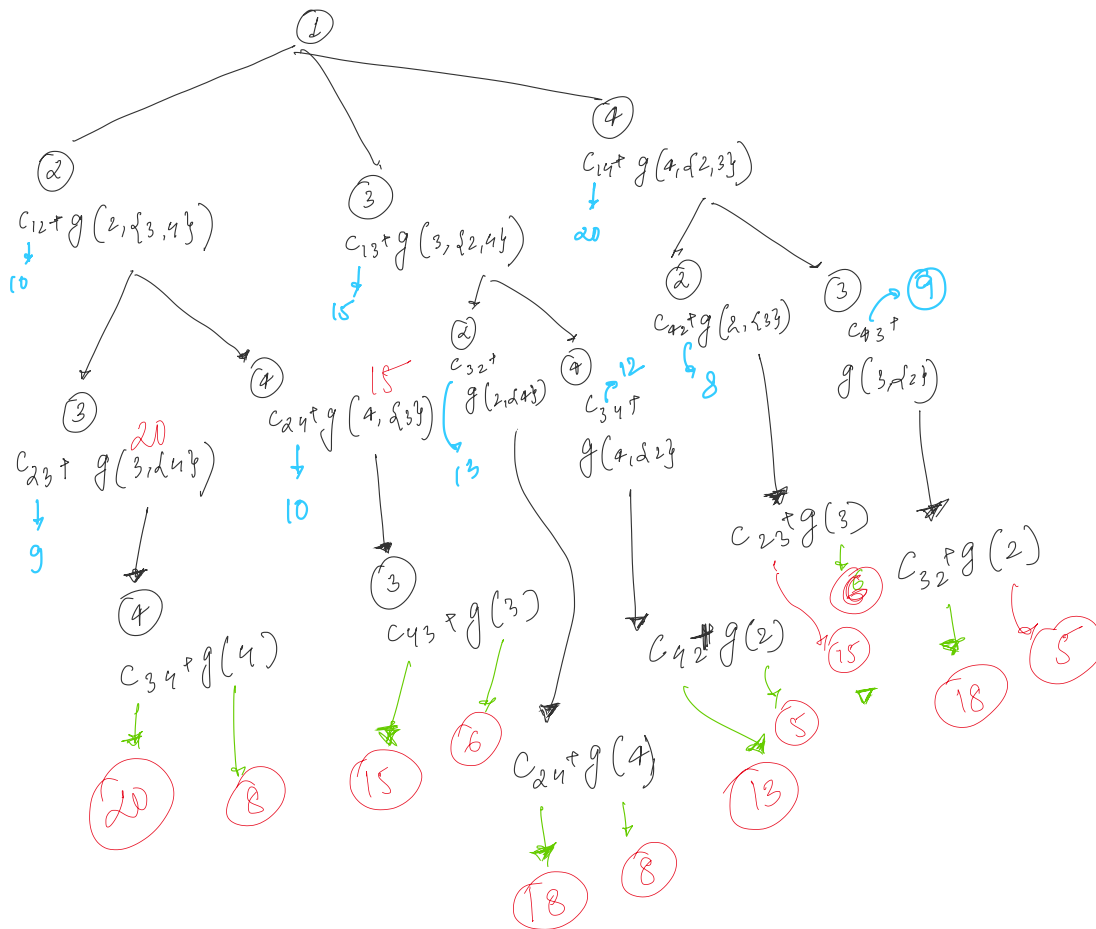


	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

Q We have to start from some vertex and travel all the other vertex once and return back to same starting vertex.
and cost of travelling must be minimum.

$$g(i, S) = \min_{k \in S} \{ c_{ik} + g(k, S - \{i\}) \}$$

$$g(1, \{2, 3, 4\}) = \min_{k \in \{2, 3, 4\}} \{ c_{1k} + g(k, \{2, 3, 4\} - \{1\}) \}$$



Other way to solve:-

* we will find out the last level value

$$g(2, \emptyset) = 5 \quad \text{last level} \quad g(2, \{2, 3, 4\}) = 25$$

$$g(3, \emptyset) = 6$$

$$g(4, \emptyset) = 8$$

$$g(2, \{3\}) = 15$$

$$g(2, \{4\}) = 18$$

$$g(3, \{2\}) = 18$$

$$g(3, \{4\}) = 20$$

$$g(4, \{2\}) = 13$$

$$g(4, \{3\}) = 15$$

second last level

$$g(3, \{2, 4\}) = 25$$

$$g(4, \{2, 3\}) = 23$$

$$g(1, \{2, 3, 4\}) = 35$$